



PAWN SHOP

Software Testing Term Project

PRESENT BY

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Introduction

- A small-scale offline system designed for a gold pawn shop business
- Replace handwritten slips and manual calculations
- Enhance efficiency, accuracy, customer satisfaction, security

System Overview

We have **3** main processes

- **Pawning Process** : Facilitates the pawning of jewelry in exchange for loans.
- **Redeeming Process** : Allows customers to reclaim their pawned items by repaying the loan amount plus any accrued interest.
- **Generating Reports** : Provides daily and monthly reports to offer insights into the shop's financial performance. (not in functional scope)

System Scope

The primary focus of this project is on **two critical functions**:

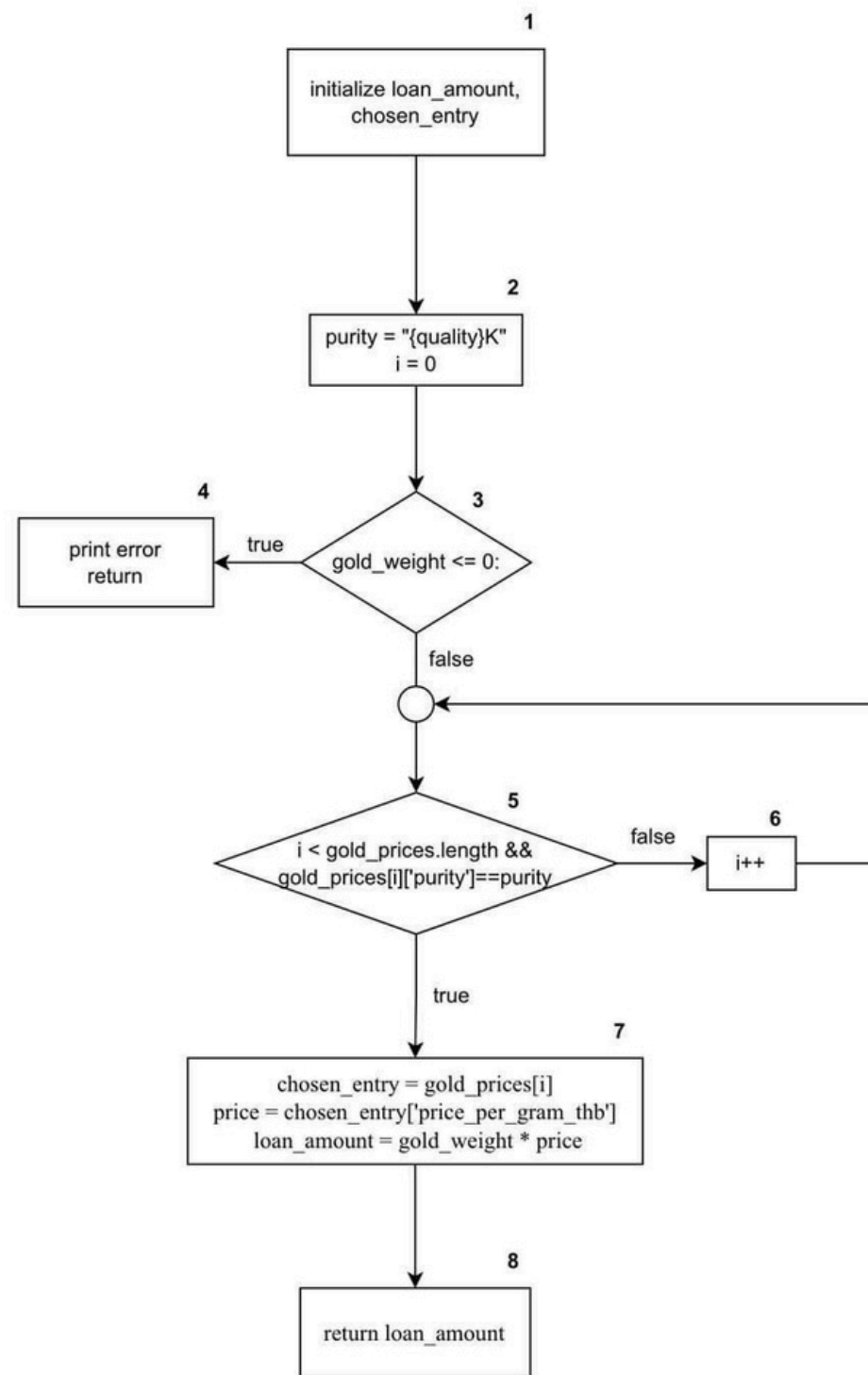
- **Loan Estimation** : Calculates the loan amount based on the weight and purity of pawned items, ensuring customers receive fair offers.
- **Interest Calculation** : Computes the total interest owed upon item redemption, providing transparency in repayment.

PSEUDO-CODE FOR THE LOAN ESTIMATION FUNCTION

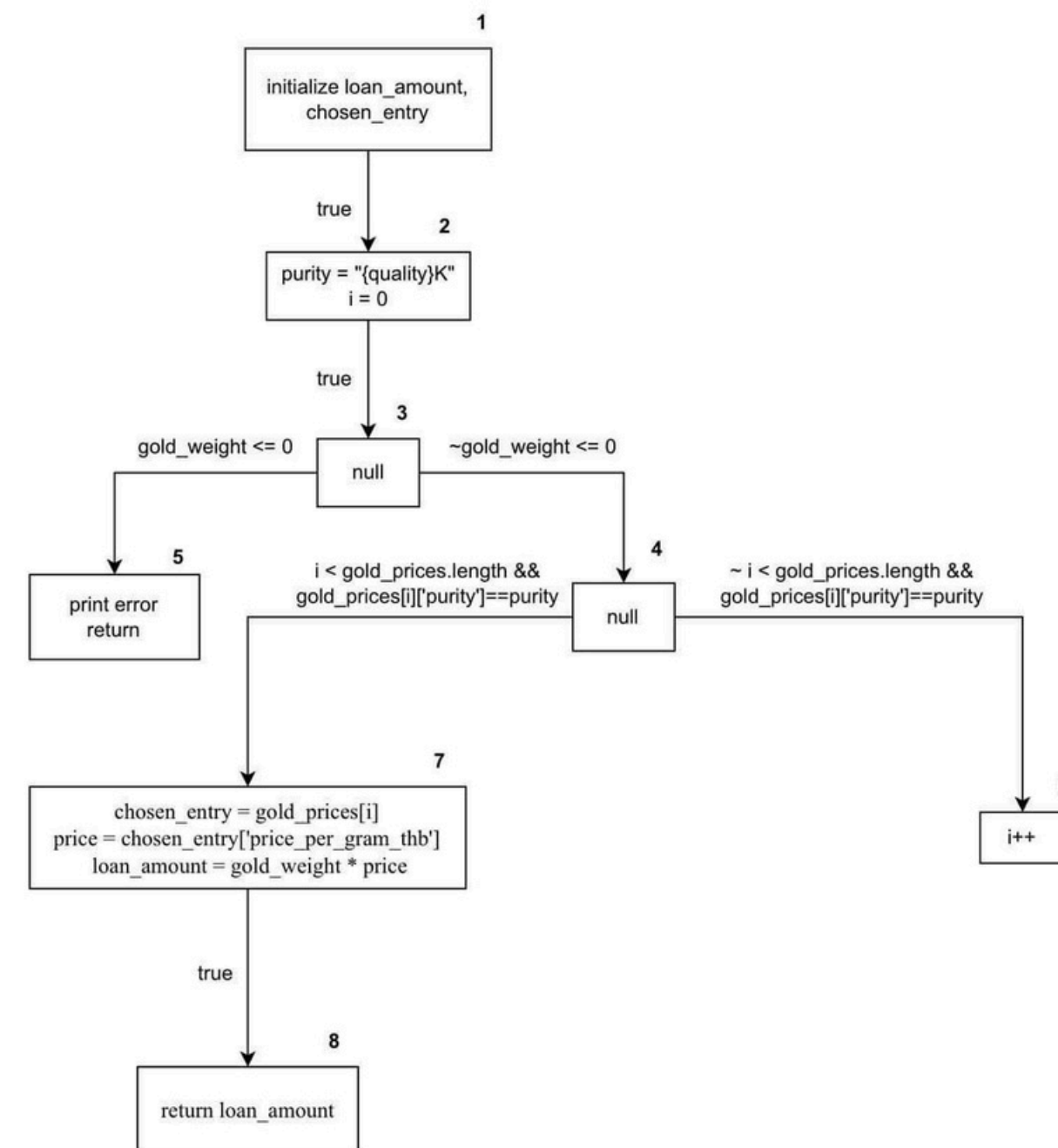
```
def calculate_loan_estimation(gold_weight, gold_prices, quality):  
    initialize price_entry, price, loan_amount;  
    purity = f"{quality}K"  
  
    chosen_entry = None  
    for price_entry in gold_prices:  
        if price_entry['purity'] == purity:  
            chosen_entry = price_entry  
            break  
  
    price = chosen_entry['price_per_gram_thb']  
  
    if gold_weight ≤ 0:  
        raise ValueError("Gold weight must be greater than zero.")  
  
    loan_amount = gold_weight * price  
  
    return loan_amount
```

- Allows users to estimate the loan amount based on the jewelry's weight (in grams) and its purity level.

CFG AND DFG FOR THE LOAN ESTIMATION FUNCTION



Control Flow Graph for Loan Estimation



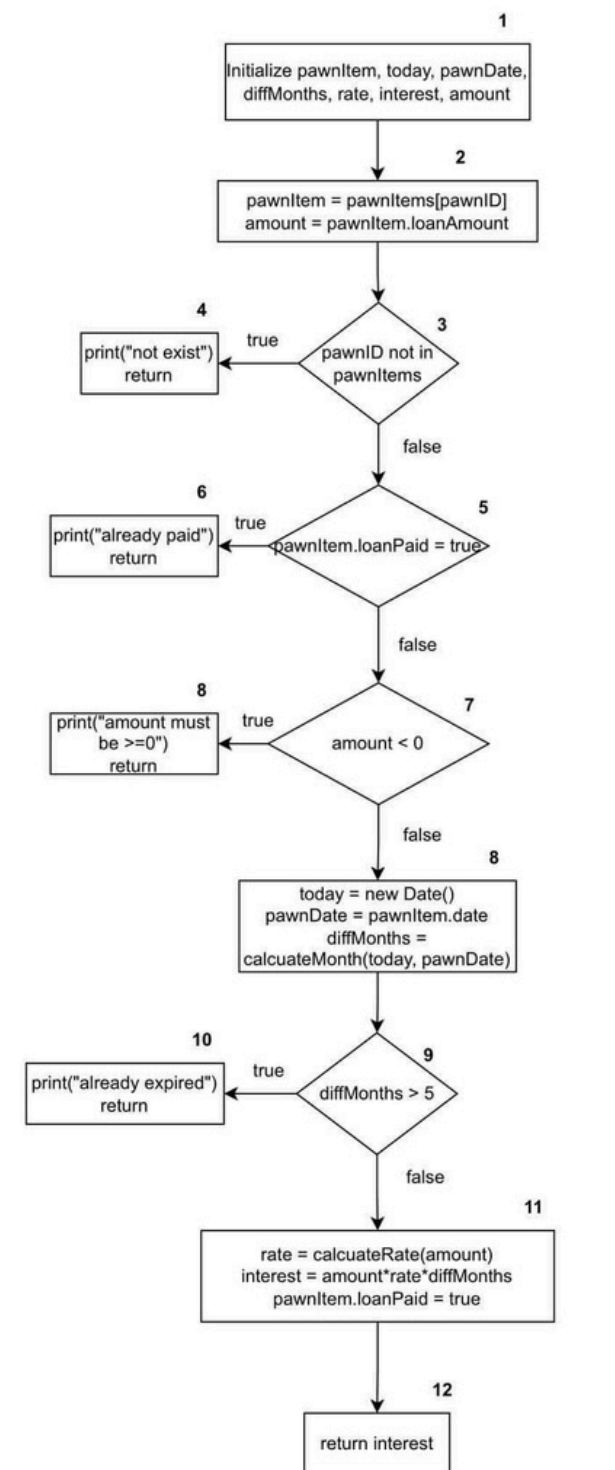
Data Flow Graph for Loan Estimation

PSEUDO-CODE FOR THE INTEREST CALCULATION FUNCTION.

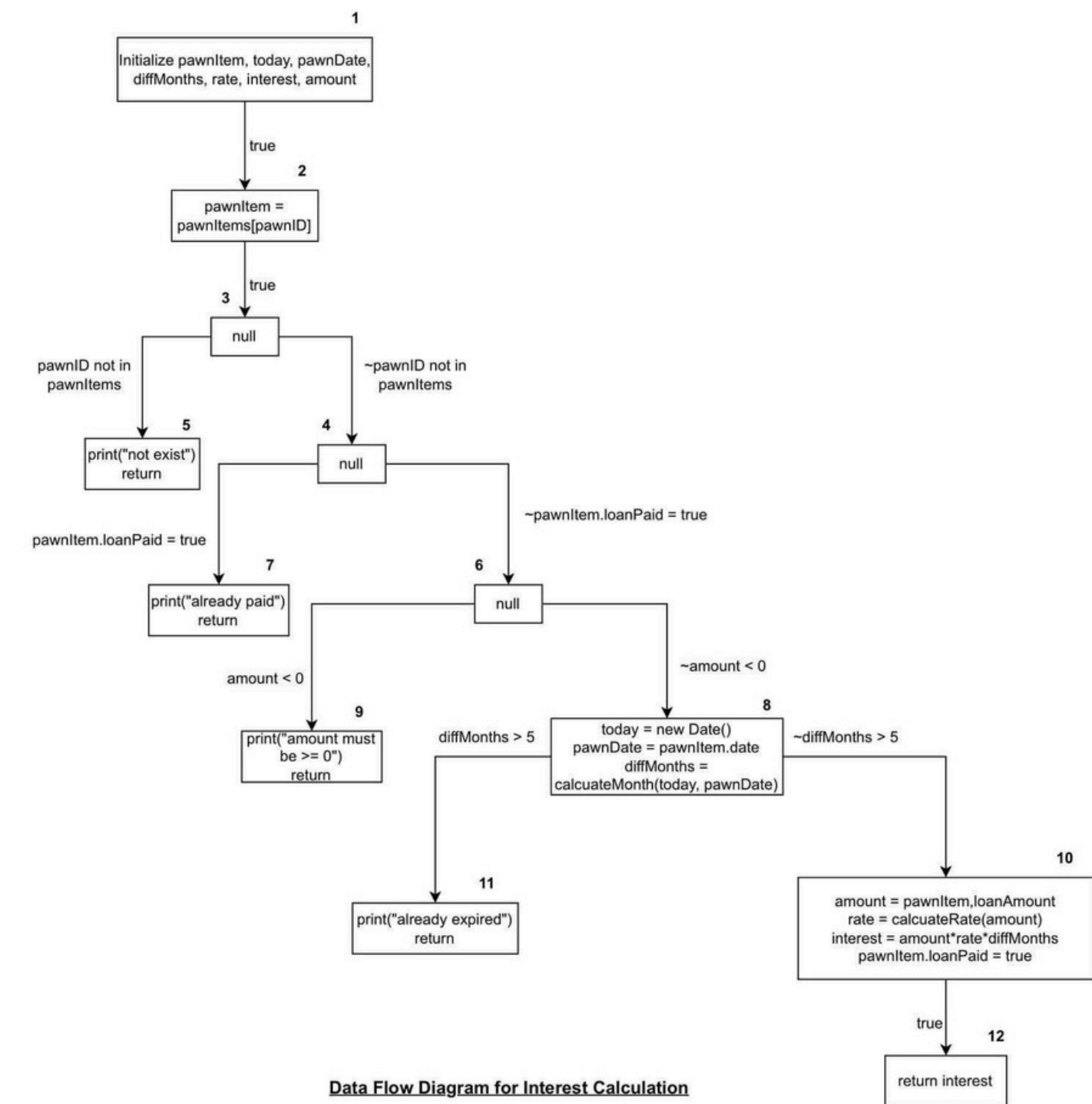
```
function calculateInterest(pawnID, pawnItems):  
    pawnItem pawnItem  
    Date today, pawnDate  
    Double diffMonths, rate, interest;  
  
    pawnItem = pawnItems[pawnID]  
    amount = pawnItem.loanAmount  
  
    if pawnItem not in pawnItems:  
        print("Pawn ID does not exist")  
        return  
  
    if pawnItem.loanPaid:  
        print("Loan is already paid")  
        return  
  
    if amount < 0:  
        print("Loan Amount must be greater than zero")  
        return  
  
    today = new Date()  
    pawnDate = new Date(pawnItem.date)  
    diffMonths = calculateMonth(today, pawnDate)  
  
    if diffMonths > 5:  
        print("Loan period is longer than 5 months")  
        return  
  
    rate = calculateRate(amount)  
    interest = amount * rate * diffMonths  
    pawnItem.loanPaid = true  
  
    return interest
```

- Calculates the total interest payable for redeeming a pawned item using the pawn slip ID and predefined rules

CFG AND DFG FOR THE INTEREST CALCULATION FUNCTION.



Control Flow Diagram for Interest Calculation



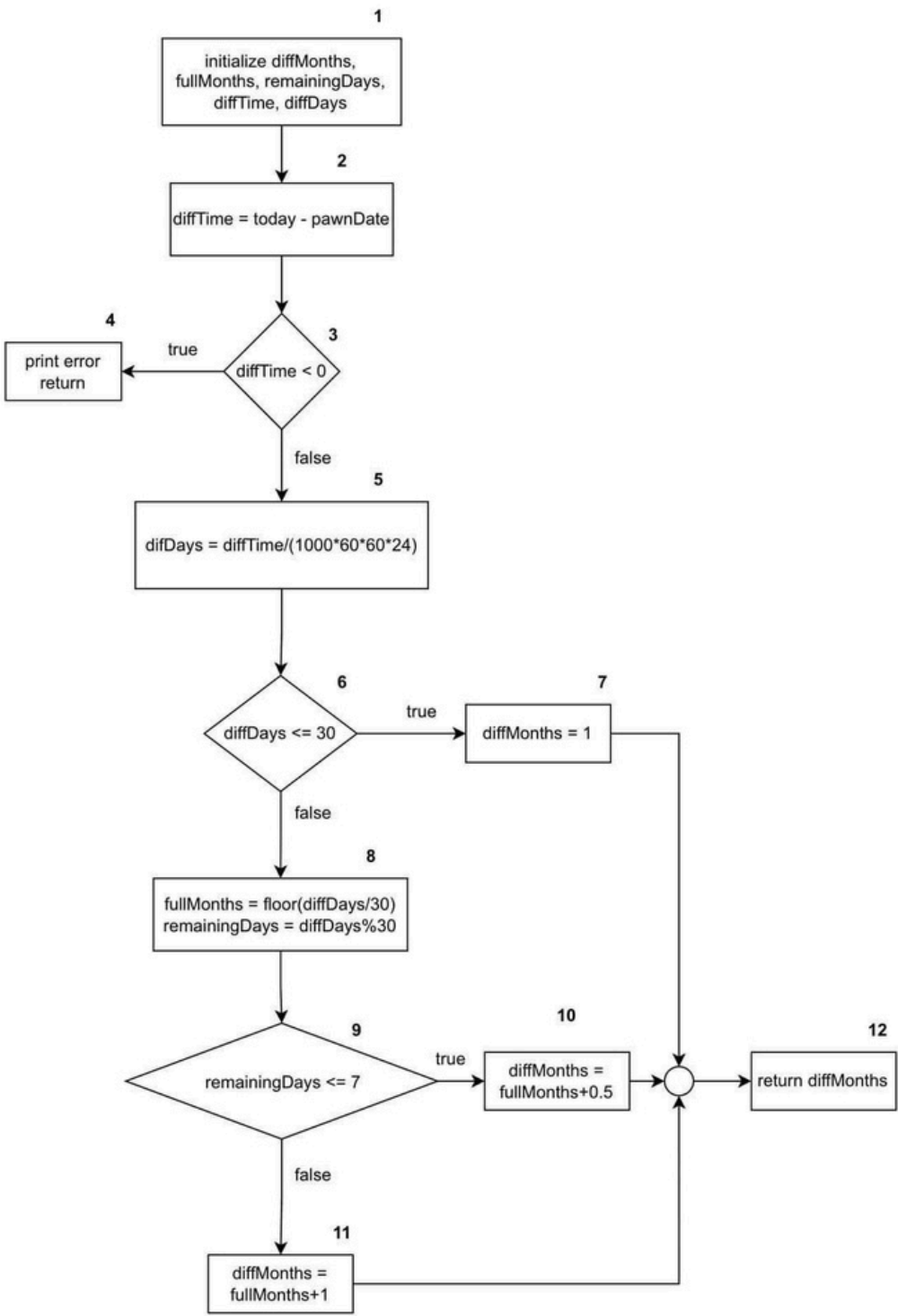
Data Flow Diagram for Interest Calculation

PSEUDO-CODE FOR CALCULATEMONTH

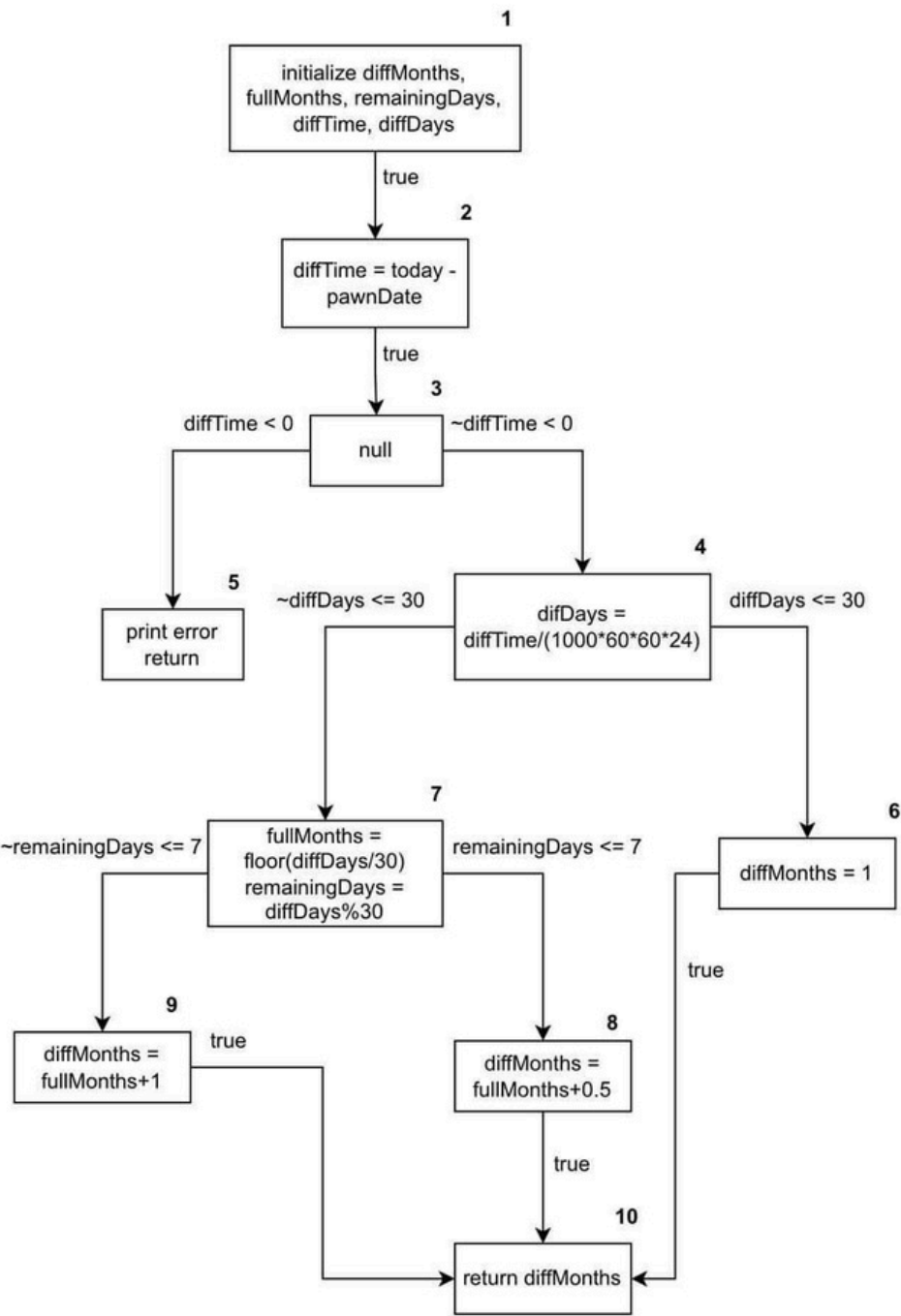
```
function calculateMonth(today, pawnDate):  
    Double diffMonths  
    Integer fullMonths, remainingDays  
    Integer diffTime, diffDays;  
  
    diffTime = today - pawnDate  
    if diffTime < 0:  
        print("Today's date must be later than the pawn date")  
        return  
  
    diffDays = diffTime / (1000 * 60 * 60 * 24)  
  
    if diffDays ≤ 30:  
        diffMonths = 1  
    else:  
        fullMonths = floor(diffDays / 30)  
        remainingDays = diffDays % 30  
        diffMonths = fullMonths  
        if remainingDays ≤ 7:  
            diffMonths += 0.5  
        else:  
            diffMonths += 1  
  
    return diffMonths
```

- Determine the period between the pawning date and the redeeming date.

CFG AND DFG FOR CALCULATEMONTH



Control Flow Diagram for Month Calculation



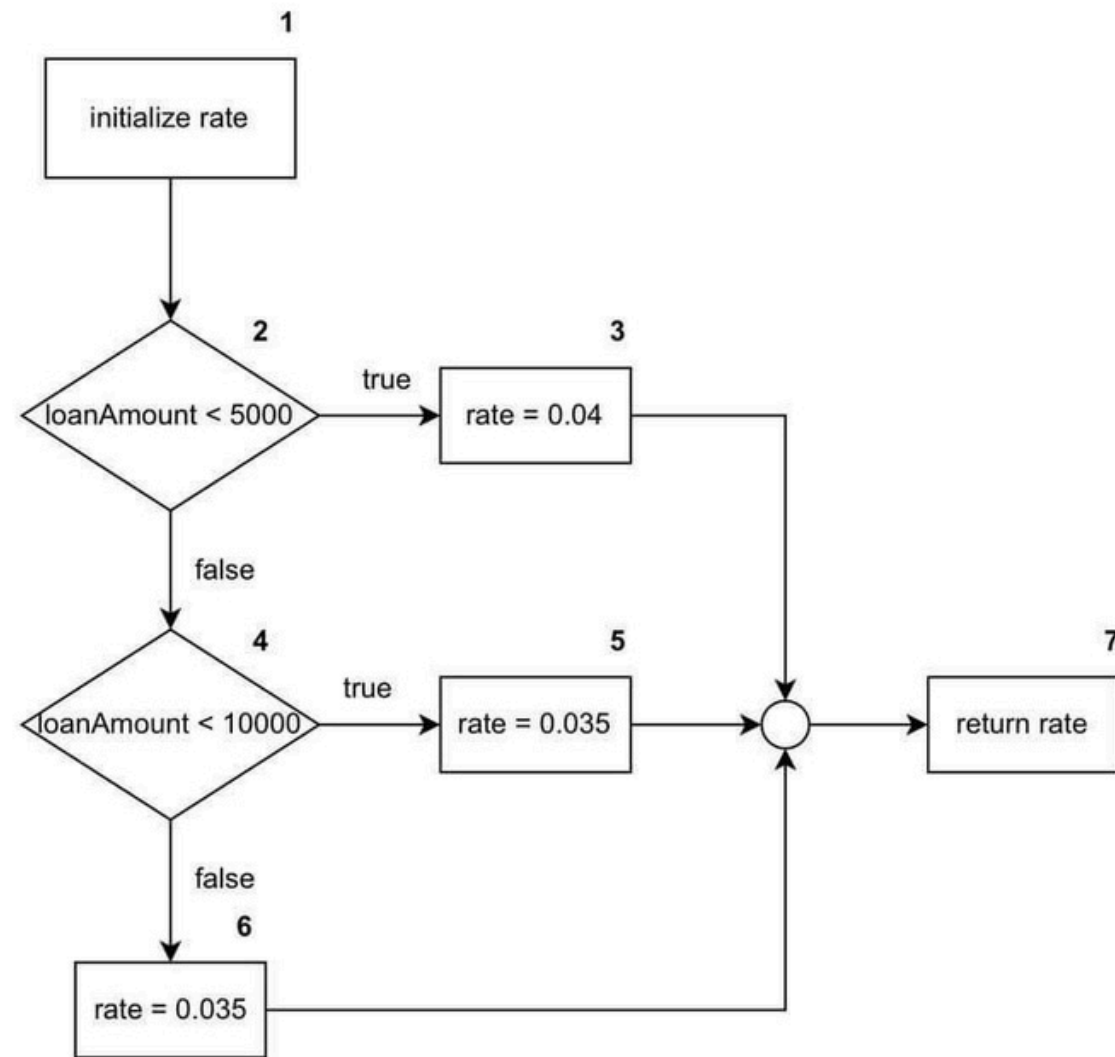
Data Flow Diagram for Month Calculation

PSEUDO-CODE FOR CALCULATERATE

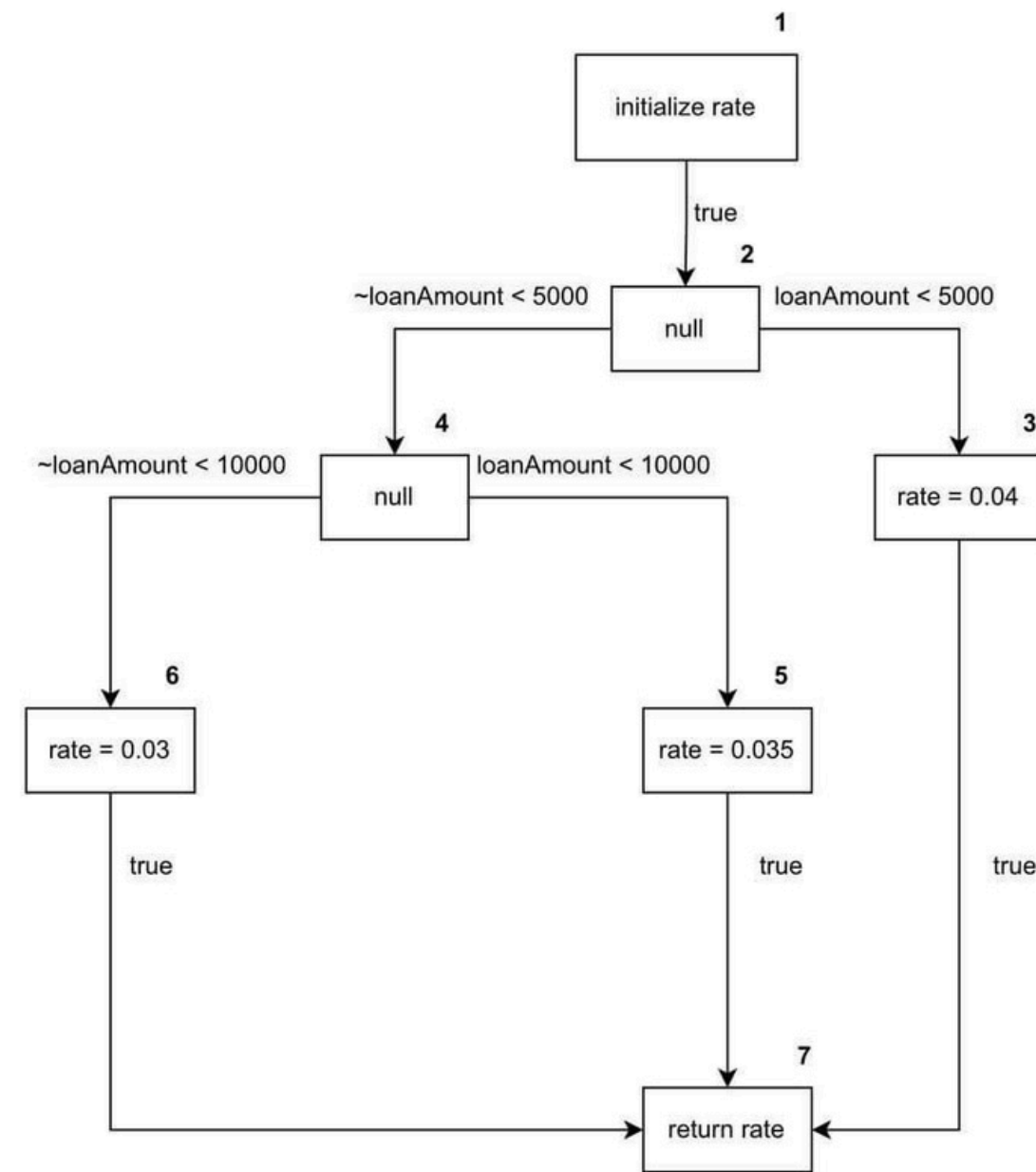
```
function calculateRate(loanAmount):  
    Double rate;  
  
    if loanAmount < 0:  
        print("Loan Amount must be greater than zero")  
        return 0  
    else if loanAmount < 5000:  
        rate = 0.04  
    else if loanAmount < 10000:  
        rate = 0.035  
    else:  
        rate = 0.03  
  
    return rate
```

- Determine the interest rate based on the loan amount.

CFG AND DFG FOR CALCULATERATE



Control Flow Diagram for Interest Rate Calculation



Data Flow Diagram for Interest Rate Calculation

How the code are tested?

Testing Strategy

- **Unit Testing:** Testing each function in isolation to ensure they performed as expected.
- **Scenario-based Testing:** Checking normal operations, edge cases, and invalid inputs to validate each function.



- **Test Planning** : Derive the normal test cases (**functional testing**) and edge test cases (**Boundary testing**)
- **Test Execution** : Run the test cases and compare with expected outputs
- **Validation** : Observe the actual outputs and expected outputs



Loan Estimation Function Testing

Valid Cases

Case 1 : Valid Loan Estimation

- TestedID : LE1
- Input: `gold\_weight = 20`, `quality = 18`
- Output: Loan is estimated as per the formula (e.g., loan = `gold\_weight` x `gold\_price`)
- Expected Output : “The shop can offer up to : 32000.00 THB.”

Case 2 : Gold Weight with Floating points

- TestedID : LE2
- Input: `gold\_weight = 13.5`, `quality = 22`
- Output: Loan is estimated as per the formula (e.g., loan = `gold\_weight` x `gold\_price`)
- Expected Output : “The shop can offer up to : 25650.00 THB.”

Loan Estimation Function Testing

Negative Cases

Case 3 : Gold Weight is Zero

- TestedID : LE3
- Input: `gold\_weight = 0`, `quality = 22`
- Output: Error message indicating that please enter a valid value.
- Expected Output : "Error: Gold weight must be greater than zero."

Case 4: Gold Weight is Negative

- TestedID : LE4
- Input: `gold\_weight = -0.01`, `gold\_quality = 24`
- Output: Error message indicating that please enter a valid value.
- Expected Output : "Error: Gold weight must be a positive number."

Interest Calculation Function Tests

Checking Existence of pawn slip

Valid Case

- TestID: IC2
- Input : pawn slip ID which exists in the database (ID - PAWN025)
- Output : Interest : 2268

Negative Case

- TestID: IC1
- Input : pawn slip ID which doesn't exist in the database
- Output : "Pawn ID does not exist"

Checking the Status of pawn slip

Valid Case

- TestID: IC4
- Input: pawn slip ID which is not paid yet (ID - PAWN025)
- Output: Interest: 2268

Negative Case

- TestID: IC3
- Input: pawn slip ID which is already paid
- Output: "Loan is already paid"

Interest Calculation Function Tests(Month Calculation)

Suppose today is 10th September 2024

Valid Case

Case 1: valid dates

- TestID: IC8
- Pawn date : 25th Aug 2024 (less than 30 days)
- Output : 1 month

- TestID: IC11
- Pawn date $\Rightarrow$ 9th May 2024(124 days)
- Output $\Rightarrow$ 4.5 months

- TestID: IC13
- Pawn date $\Rightarrow$ 22nd July 2024(50 days)
- Output $\Rightarrow$ 2 months

Negative Case

Case 2: Invalid date

- TestID: IC14
- Pawn date $\Rightarrow$ 12th April 2024(151 days)
- Output $\Rightarrow$ error: “Slip expired”

- TestID: IC16
- Pawn date $\Rightarrow$ 15th Sep 2024
- Output $\Rightarrow$ error: “The pawn date hasn't arrived yet!”

Interest Calculation Function Tests(Defining Interest Rate)

Case 1: Loan amount less than 5000 THB

- TestID: IC17
- Loan Amount: 2100 THB
- Output $\Rightarrow$ Interest Rate: 0.04

Case 2: Loan amount is more than or equal to 5000 THB but less than 10,000 THB

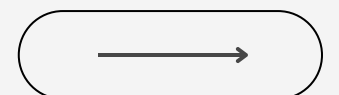
- TestID: IC18
- Loan Amount: 7400 THB
- Output $\Rightarrow$ Interest Rate: 0.035

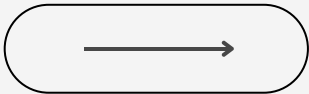
Case 3: Loan amount more than or equal to 10,000 THB

- TestID: IC19
- Loan Amount: 32,200 THB
- Output $\Rightarrow$ Interest Rate: 0.03

UI Testing

We have used simple but effective user interface to receive or give out information from our user.





[Loan Estimation](#) [Pawn Slip Registration](#) [Interest Calculation](#) [Report](#)

Purity	Price per Gram (THB)
18K	1600
22K	1900
24K	2100

Update Gold Price

Gold Quality:
18 Karat ▾

Gold Weight (grams):
10.3

Gold Price (THB per gram):
1600

Submit

The shop can offer up to:
16480.00 THB

Update Gold Price

Gold Quality:
18 Karat ▾

New Price (THB):

Update

Cancel

Gold Quality:
18 Karat ▾

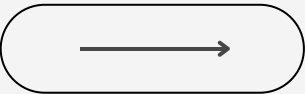
Gold Weight (grams):
-20

Please enter a valid value

Gold Price (THB per gram):
1600

Submit

Loan estimation page UI elements



Loan Estimation

Pawn Slip Registration

Interest Calculation

Report

Interest Calculation

Pawn ID:

PAWN034

Calculate

Loan Amount: 21600 THB

Interest Rate:

0.03

Months: 2

Total Interest: 1296.00 THB

Total Payment: 22896 THB

Loan Paid

Interest Calculation

Pawn ID:

PAWN022

Calculate

Pawn ID does not exist

Interest Calculation

Pawn ID:

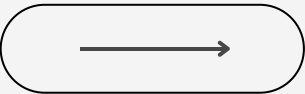
PAWN025

Calculate

Loan is already paid

Interest Calculation Page

Report page (not in functional scope)



Loan Estimation

Pawn Slip Registration

Interest Calculation

Report

Daily Report

Total Loans Amount (THB)	Total Interest (THB)	Loans Left	Total Number of Clients	Total Profit (THB)	Bad Debt (THB)
150,000.00	5,000.00	3	7	7500.00	2,500.00

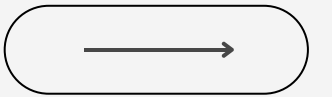
Weekly Report

Total Loans Amount (THB)	Total Interest (THB)	Loans Left	Total Number of Clients	Total Profit (THB)	Bad Debt (THB)
1,000,000.00	35,000.00	15	45	50,000.00	15,000.00

Monthly Report

Total Loans Amount (THB)	Total Interest (THB)	Loans Left	Total Number of Clients	Total Profit (THB)	Bad Debt (THB)
4,200,000.00	150,000.00	60	180	200,000.00	50,000.00

Pawn Slip Registration page (not in our functional scope)



[Loan Estimation](#) [Pawn Slip Registration](#) [Interest Calculation](#) [Report](#)

Pawn Slip Registration

Pawn ID:

Gold Amount:

Purity:

Date:

Customer Name:

City:

Conclusion

Software Testing process for Loan Estimation and Interest Calculation ensures that our system is fail proof while also having an easy interface to learn

link to deployment : <https://reon1917.github.io/pawn-shop-app/>

